

Viewer Retention Recommendations

Prioritization Matrix, Risk Assessment & A/B Testing Framework

OTT Viewer Drop-off & Retention Analysis — Phase D

All recommendations below are directly linked to statistically validated findings from Phase B (Root Cause Analysis) and Phase C (Segmentation).

1 2×2 Impact–Feasibility Matrix

Each recommendation is plotted on impact (effect on drop-off / retention) versus feasibility (ease and speed of implementation). Quick Wins are implemented first; Strategic Bets require phased planning; Fill-ins are pursued if bandwidth allows.

	High Feasibility	Low Feasibility
High Impact	QUICK WINS R1: Smarter Autoplay & Next-Episode UX R2: Episode-8 Recap / Re-entry Nudges R3: Pacing-Recovery Sequencing Rule	STRATEGIC BETS R5: Script Guidelines (Hook + Pacing + Cognitive Load) R6: Genre Diversification for High-Drama Platforms
Medium Impact	FILL-INS R4: Pause-Triggered “Interrupted Viewer” Recap Prompt R7: Hide “Skip Intro” on Episode 1 for First-Time Viewers	DEPRIORITIZE <i>(none identified — all recommendations are grounded in validated drivers)</i>

2 Recommendation 1: Smarter Autoplay & Next-Episode UX for At-Risk Episodes

R1 — Quick Win

Data Insight: The pre-air content risk score (combining `pacing_score`, `hook_strength`, and `cognitive_load`) reliably flags episodes likely to cause drop-off *before* the next-episode transition occurs. Episodes scoring in the high-risk band show 7–23× higher drop-off than low-risk episodes at the same season position.

Action: For episodes flagged high-risk by the content risk score, modify the post-episode autoplay experience: extend the “what’s coming next” teaser, surface a stronger hook for the upcoming episode in the next-episode card, or adjust the autoplay countdown timing

specifically for these transitions (rather than a one-size-fits-all autoplay rule).

Impact: High — intervenes at the exact point of viewer departure (the episode-to-episode transition), where the data shows continuation rate (CR) drops most sharply for high-risk episodes.

Feasibility: High — platform-side UX/UI feature using existing risk-scoring infrastructure; no content rework required; can be deployed incrementally per title.

Risk: Over-engineering autoplay (e.g., overly aggressive teasers or forced countdowns) can feel manipulative, reducing trust and organic viewing choice. Viewers may perceive the platform as “pushing” content.

Mitigation: Apply the modified UX *only* to episodes flagged high-risk (not platform-wide); A/B test multiple UX variants against a standard-autoplay control; always provide an easy “skip” / opt-out so the intervention augments rather than overrides viewer choice.

Primary KPI: $CR_{k \rightarrow k+1}$ (continuation rate) for episodes flagged high-risk, measured before vs. after the UX change.

Secondary KPIs: Season completion rate (CR_{season}); autoplay engagement rate (% who let autoplay proceed vs. manually navigate); opt-out / skip rate of the new UX element.

3 Recommendation 2: Episode-Level Recaps & Re-entry Nudges at Key Confusion Points (Especially Episode 8)

R2 — Quick Win

Data Insight: The episode-position $\times$ pacing heatmap identified episode 8 combined with `pacing_score = 3` as the single worst cell in the dataset, with a 71.0% drop-off rate — by far the most concentrated risk pocket found in the entire season arc. More broadly, slow-paced episodes maintain 40–55% drop-off at every position, but episode 8 is where this risk peaks most sharply.

Action: Insert a short “Previously on...” recap card or re-entry nudge immediately before identified high-risk episodes (especially episode 8 and similar danger-zone positions flagged by the risk heatmap). Personalize the recap trigger based on the viewing gap — only surface it if time elapsed since the previous episode exceeds a threshold (e.g., 48 hours), so binge-watchers are not interrupted unnecessarily.

Impact: High — targets a precisely identified, statistically extreme danger zone (71% drop-off) rather than a diffuse intervention; even a modest relative reduction at this single point yields a large absolute improvement in season completion.

Feasibility: High — recap cards are a standard, templatable platform feature; can be rolled out per-show using existing metadata (episode synopsis) without new content production.

Risk: Recaps may feel redundant or patronizing to viewers who are actively engaged and remember the plot, potentially increasing skip-behavior (mirroring the `skip_intro` pattern, which itself correlates with a 4 $\times$ higher drop-off risk) or mild irritation.

Mitigation: Make the recap fully skippable with a single tap; gate it behind the viewing-gap personalization rule so it only appears for viewers likely to need it; A/B test recap presence vs. absence specifically at episode 8 to confirm a net positive effect before extending to other flagged positions.

Primary KPI: Drop-off rate at episode 8 (and other flagged danger-zone positions), targeting a measurable reduction from the 71.0% baseline in the worst-case cell.

Secondary KPIs: avg_watch_percentage at the targeted episode positions; recap skip/dismiss rate; re-entry completion rate (% who finish the episode after viewing the recap).

4 Recommendation 3: Mandatory Pacing-Recovery Sequencing Rule

R3 — Quick Win

Data Insight: Pacing sequence transition analysis shows that a Slow episode following another Slow episode produces ~42.7% drop-off — statistically indistinguishable from a Fast→Slow “whiplash” transition (~42.9%). However, a Slow episode immediately followed by a Fast episode drops to just ~2.5% drop-off. The *sequence*, not any single episode’s pacing in isolation, determines the outcome.

Action: At the season-outlining stage, enforce a rule: no two consecutive episodes may both carry a `pacing_score` ≤ 4 . If a slow-burn episode is creatively necessary, the following episode must score ≥ 6 on pacing.

Impact: High — a single ordering rule can rescue a large share of the “sustained slow” episode pool (currently 40–55% drop-off in that pacing tier) without altering any individual episode’s content.

Feasibility: High — this is a scheduling/ordering decision made during outlining, before any episode is written or shot. No reshoots or new content required.

Risk: Forcing a “fast” episode after every slow one could feel formulaic, and may conflict with narrative arcs that intentionally build tension across multiple slow episodes (e.g., mystery/drama slow-burns).

Mitigation: Define “fast” flexibly — `pacing_score` ≥ 6 can be achieved via tighter editing, converging subplots, or a strong cliffhanger, not just action. Pilot on 3–5 new productions before mandating platform-wide, with a documented creative-exception process for justified consecutive slow episodes.

Primary KPI: $CR_{k \rightarrow k+1}$ for episodes immediately following a slow-paced (`pacing_score` ≤ 4) episode, targeting movement from ~57% (sustained-slow continuation) toward the ~97.5% seen in Slow→Fast transitions.

Secondary KPIs: % of season episode-pairs compliant with the sequencing rule; overall season drop-off rate; writers’-room compliance/exception rate.

5 Recommendation 4: Real-Time “Interrupted Viewer” Intervention (Pause-Triggered Recap Prompt)

R4 — Fill-In

Data Insight: The behavioral profile analysis identified an “Interrupted (High Pauses)” viewer segment — episodes where `pause_count` ≥ 6 — carrying 39.7% drop-off, nearly 3× the 14.5% baseline. `pause_count` shows a meaningful positive correlation with drop-off ($r \approx +0.31$) and is detectable in real time during playback.

Action: When a viewer’s pause count crosses a threshold (e.g., 5+) within a single episode, surface a lightweight, dismissible “Need a quick recap?” prompt during playback, or queue a “Continue where you left off” nudge for their next session.

Impact: Medium — this is a real-time rescue mechanism for viewers already showing distress signals, addressing the consequence rather than the underlying content cause; affects a meaningful but smaller population than R1–R3.

Feasibility: High — purely a platform/product feature using telemetry (`pause_count`) already being tracked; no content team involvement; cheap to A/B test.

Risk: Viewers may pause for legitimate non-confusion reasons (interruptions, phone calls), so the prompt could misfire and feel intrusive, risking “prompt fatigue” and a negative reaction to the intervention itself.

Mitigation: Set the threshold conservatively above the dataset average (mean `pause_count` ≈ 3.1 , so trigger at 5+); make the prompt one-tap dismissible and limit to once per episode; A/B test against a silent control to confirm it improves — rather than harms — next-session return rate before full rollout.

Primary KPI: Drop-off rate within the “Interrupted” behavioral profile segment, targeting reduction from the 39.7% baseline.

Secondary KPIs: Prompt engagement/dismiss rate; next-session return rate for viewers who received the prompt vs. control.

6 Recommendation 5: Script & Content Guidelines for Mid-Season Pacing, Hooks and Cliffhangers

R5 — Strategic Bet

Data Insight: Three validated root causes converge here: `hook_strength` shows a 7–23× drop-off risk multiplier between low (3–5) and high (8–10) tiers at *every* episode position; `cognitive_load` is the single strongest overall predictor of drop-off ($r \approx +0.57$); and the “Prestige Tension” segment shows that high cognitive load is tolerable (~12–18% drop-off) *only when paired with a strong hook*, whereas weak hook + high cognitive load + slow pacing defines the “Burnout Zone” (~40% drop-off, ~43% avg. watch).

Action: Establish formal script-stage guidelines for mid-season episodes (positions 4–10, where structural “ep1 launch effect” no longer applies): (a) `hook_strength` ≥ 6 required at script review, with sub-threshold scripts routed to consultation rather than rejected outright; (b) high cognitive load (≥ 7) permitted only if `hook_strength` ≥ 7 on the same episode; (c) cliffhanger strength explicitly assessed alongside pacing for episodes preceding

a planned slow-burn.

Impact: High — attacks the two strongest content-design root causes simultaneously, specifically in the mid-season window where most content-driven (non-structural) drop-off occurs.

Feasibility: Low — requires coordinated script-stage intervention across hook design and narrative complexity on real productions, creative-team buy-in, and calibrated scoring rubrics; cannot be deployed as a quick platform toggle.

Risk: (1) Hook_strength scoring is subjective and could be gamed by writers seeking to clear the threshold. (2) Capping unsupported high-cognitive-load episodes may be perceived as “dumbing down” prestige storytelling, alienating the audience segment that values complexity.

Mitigation: Run a scoring-calibration exercise across multiple reviewers before roll-out. Frame the guideline explicitly: high cognitive load is *not* restricted — only the *unsupported combination* (high cognitive load + weak hook) is flagged, using the Prestige Tension vs. Burnout Zone gap (12–18% vs. 40% drop-off) as the evidentiary basis communicated to writers’ rooms.

Primary KPI: Episode-level drop-off rate for mid-season episodes (positions 4–10), targeting movement away from the Burnout Zone profile (40% drop-off) toward the Prestige Tension profile (12–18%) or better.

Secondary KPIs: Average hook_strength and pacing_score at script-review stage (pre/post guideline introduction); % of episodes meeting both thresholds; reviewer-calibration agreement score.

7 Recommendation 6: Genre Diversification for High-Drama Platforms

R6 — Strategic Bet

Data Insight: Drama carries the platform’s highest drop-off rate at 24.2% — $1.7\times$ the 14.5% overall baseline — driven by its content DNA (66.3% high-cognitive-load episodes, 49.8% slow-paced episodes). By contrast, Comedy shows only 2.9% drop-off with 0% slow-paced episodes and 25.4% high-cognitive-load episodes, demonstrating that a different content mix structurally produces far better retention outcomes.

Action: For platforms or content slates with a disproportionately high share of Drama titles, gradually diversify the new-commission mix to include a higher proportion of genres with structurally favorable retention profiles (e.g., Comedy, Action & Adventure), without abandoning Drama as a core offering.

Impact: High — shifting the platform-level genre mix directly shifts the platform-wide drop-off composition, since Drama is both high-volume and high-drop-off relative to baseline.

Feasibility: Low — requires multi-season content investment, slate planning, and commissioning lead times of 12+ months; results are not observable in the short term and carry significant financial commitment.

Risk: Diluting a platform’s Drama-led brand identity risks alienating its core/loyal

Drama audience; large upfront content investment with delayed and uncertain payoff; genre diversification could underperform if new titles are poorly executed.

Mitigation: Pursue gradual diversification (a small % of each new commissioning cycle) rather than an abrupt shift; pilot with a limited regional or platform-segment rollout; monitor brand-sentiment metrics alongside drop-off so retention gains are not offset by brand-perception losses. Combine with R5 so that any new Drama titles still commissioned benefit from improved content design standards.

Primary KPI: Platform-level overall drop-off rate, tracked over multiple commissioning cycles as genre mix shifts.

Secondary KPIs: Genre-mix ratio (% Drama vs. other genres in active slate); new-genre title performance vs. the Drama benchmark (24.2%) and platform baseline (14.5%); subscriber churn rate; brand-sentiment score.

8 Recommendation 7: Hide “Skip Intro” on Episode 1 for First-Time Viewers

R7 — Fill-In

Data Insight: `skip_intro = 1` correlates with a 23.3% drop-off rate versus 5.9% for non-skippers — a 4× risk multiplier, one of the strongest behavioral signals identified. Separately, episode 1 carries the platform’s *lowest* structural drop-off (3.7% vs. the 14.5% baseline), reflecting a “launch curiosity effect” where new viewers are maximally attentive. Hook_strength — whose effect is established largely through the cold open / intro sequence — shows a 7–23× drop-off risk multiplier between weak and strong tiers at *every* episode position in the season. For a first-time viewer, skipping this context-setting sequence before any investment in the show has formed may undercut the hook’s intended effect for the remainder of the season.

Action: For first-time viewers of a series (identified via no prior watch history for that `show_id`), hide the dedicated “Skip Intro” button specifically on Episode 1, while still allowing manual scrubbing/seeking through the intro via the playback seek bar. From Episode 2 onward, and for all returning viewers, “Skip Intro” remains available as normal.

Impact: Medium — the intervention is deliberately narrow in scope (Episode 1, first-time viewers only), but it targets the highest-leverage moment for hook formation identified in the data; effects should be measured primarily through downstream Episode 2 continuation rather than Episode 1 alone.

Feasibility: High — a conditional UI rule based on existing viewer-history data (first-time vs. returning × episode number); no content production changes required; can be deployed as a configuration toggle.

Risk: Viewers familiar with a franchise (e.g., spin-offs, viewers who already watched a trailer) may feel mildly restricted by the missing button; removing a familiar UI affordance — even on a single episode — could be perceived as the platform restricting viewer choice, generating negative feedback or complaints.

Mitigation: Preserve manual scrubbing/seeking through the intro at all times, so viewers who truly want to bypass it retain that ability — only the dedicated one-tap “Skip Intro” affordance is removed, not the capability. Apply the rule strictly to verified first-time viewers and Episode 1 only (never Episode 2+). A/B test to confirm the restriction itself does not increase early abandonment (e.g., exits during the intro) before any wider rollout.

Primary KPI: `avg_watch_percentage` and `drop_off` rate for Episode 1 among first-time viewers, comparing treatment (button hidden) vs. control (button shown as normal).

Secondary KPIs: `CR1→2` (Episode 1 to Episode 2 continuation rate) for first-time viewers, to test whether protecting hook exposure on Episode 1 improves downstream continuation; `skip_intro` rate in Episodes 2+ for the same cohort (to check whether the effect persists or reverts once the button reappears); manual scrub-through-intro rate during Episode 1 (to gauge how many viewers still bypass the intro via scrubbing despite the hidden button).

8.1 General Framework Principles

- **Unit of randomization:** User-level, not episode-level. Randomizing at the episode level for the same user violates the Stable Unit Treatment Value Assumption (SUTVA) — a user’s experience of episode 8 is not independent of their experience of episode 7. All of R1, R2,

R4, and R7 should randomize entire user cohorts into treatment/control.

- **Sample size:** Determine via power analysis (e.g., `statsmodels.stats.power`) using the relevant baseline rate as p_1 and the target rate as p_2 , at 80% power and $\alpha = 0.05$. For R2, the baseline is the episode-8 drop-off rate in the affected pacing tier (71.0%); for R3, the baseline is the sustained-slow continuation rate ($\sim 57\%$).
- **Duration:** Run for a minimum of 2–3 weeks to account for novelty effects, and ideally cover at least one full release cycle so the full season arc (including the targeted episode positions) is observed for each user.
- **Segment-specific testing:** Where an intervention targets a specific segment (e.g., R5 targets Drama; R2 targets episode 8 specifically), restrict the test population to that segment to avoid diluting the effect size with unaffected users.
- **Guardrail metrics:** For every test, monitor at least one guardrail metric that would catch unintended harm — e.g., overall session-start rate (to catch UX fatigue from R1/R2/R4) or content-team satisfaction (to catch creative friction from R5).

8.2 Recommendation-Specific A/B Test Designs

Recom.	Treatment vs. Control	Validation Criterion
R1 Smarter Auto-play/UX	Control: standard autoplay. Treatment: modified next-episode UX shown only on episodes flagged high-risk by the content risk score.	Treatment group shows higher $CR_{k \rightarrow k+1}$ on flagged episodes than control, with no significant drop in overall session-start rate (guardrail).
R2 Episode-8 Recap/Re-entry Nudge	Control: no recap shown. Treatment: recap card shown before episode 8 (and other flagged positions) for users with viewing gap $> 48h$.	Treatment group's drop-off rate at episode 8 is significantly lower than control's 71% baseline (in the high-risk pacing tier), with recap skip rate $<$ guardrail threshold (e.g. $< 50\%$).
R3 Pacing-Recovery Sequencing	Control: existing/legacy show ordering. Treatment: pilot shows produced under the new sequencing rule (3–5 new productions).	$CR_{k \rightarrow k+1}$ following slow episodes in treatment shows moves toward the $\sim 97.5\%$ Slow $\rightarrow$ Fast benchmark, vs. the $\sim 57\%$ sustained-slow benchmark in control shows.
R4 Pause-Triggered Recap Prompt	Control: no prompt. Treatment: prompt shown when $pause_count \geq 5$ within an episode, one-tap dismissible.	Treatment group's drop-off rate within the "Interrupted" profile is significantly lower than control's 39.7% baseline; next-session return rate is non-inferior or better.

R5 Script Guidelines (Hook + Cognitive Load)	Control: existing Drama productions (matched on genre sub-type, platform, release year). Treatment: 2–3 pilot Drama productions following the new script guidelines.	Pilot shows' drop-off rate moves from the 24.2% Drama baseline toward the 12–18% Prestige Tension range, measured over a full season vs. matched controls; pre-defined success threshold ≥ 5 percentage-point reduction.
R6 Genre Diversification	Control: platform/region continuing current genre-mix commissioning. Treatment: platform/region piloting an increased non-Drama commissioning share.	Treatment region's platform-level drop-off rate trends below the 14.5% baseline over multiple commissioning cycles, with subscriber churn and brand-sentiment guardrails holding steady or improving.
R7 Hide Skip-Intro on Ep. 1 (First-Time Viewers)	Control: "Skip Intro" button shown as normal on Episode 1. Treatment: button hidden on Episode 1 for verified first-time viewers; manual scrubbing remains available.	Treatment group shows higher avg_watch_percentage / lower drop-off on Episode 1 and higher $CR_{1 \rightarrow 2}$ than control's 23.3% skip-associated drop-off baseline, with no increase in early-intro abandonment (guardrail).

8.3 Rollout Sequencing

Given the matrix in Section 1, the suggested validation sequence is:

1. **Immediate A/B tests (Quick Wins):** R1, R2, and R3 can be tested concurrently on different platform segments since they target distinct mechanisms (UX, recap content, and production scheduling respectively) with minimal interaction risk.
2. **Parallel Fill-in tests:** R4 and R7 can run alongside the Quick Wins as they are platform-side and low-cost, but should be monitored for interaction effects with R1/R2 and with each other — R4 (pause-triggered recap) and R7 (hidden skip-intro on Ep. 1) both modify the playback UI for new/at-risk viewers, so stacking multiple unfamiliar UI changes for the same first-time viewer could compound novelty friction. Consider testing R7 on a separate user cohort from R4 initially.
3. **Staged Strategic Bets:** R5 (script guidelines pilot) should begin production planning immediately so pilot episodes are ready once R1–R4/R7 results inform any adjustments to risk-scoring thresholds used in R1/R2. R6 (genre diversification) should only proceed after R5's pilot results are available, since R5 directly affects whether continued Drama investment is viable without diversification.